Thermal and Electrical Conductivity of Metals

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***Abstract*—This experiment determines (i) the thermal conductivity of a copper rod and (ii) the electrical conductivity of an aluminium rod. All numerical values in the report are reproducible from the data-analysis pipeline implemented in the Jupyter notebook `Analysis/calculate.ipynb`, which performs unit-consistent calculations using `pint`, loads raw data from CSV files, and estimates heat-flow rates using linear regression of time-series heat data (`scipy.stats.linregress`). The theoretical discussion is limited to the physically required relations used in the analysis (Fourier conduction and Ohm's law) and intentionally avoids unrelated or physically meaningless combinations (e.g., evaluating the Wiedemann–Franz relation using two different metals).**

I. Introduction

In insulating and non-metallic solids, heat is conducted primarily by phonons, whereas in metals heat conduction is strongly influenced by mobile electrons [1]. The purpose of this experiment is to determine the thermal conductivity of copper (λ_{Cu}) and the electrical conductivity of aluminium (σ_{Al}) from laboratory measurements.

II. Theory

Thermal conduction in a material driven by a temperature gradient is governed by Fourier's law, where the thermal flux density $\vec{J}$ is proportional to the temperature gradient:

$$\vec{J} = -\lambda \nabla T \tag{1}$$

where λ is the thermal conductivity of the material. In a one-dimensional case along a metal rod of length l and cross-sectional area A, the rate of heat transport is given by:

$$\frac{dQ}{dt} = -\lambda A \frac{\partial T}{\partial x} \tag{2}$$

For a rod that has reached steady state, the temperature profile does not change with time, i.e. $\partial T/\partial t = 0$. Under steady conditions and for a uniform rod, the spatial temperature gradient becomes approximately constant over the probe spacing, so that the magnitude of the gradient can be approximated by

$$\left|\frac{\partial T}{\partial x}\right| \approx \frac{\Delta T_{\text{rod}}}{\Delta x}. \tag{3}$$

In this report, ΔT_{rod} is taken from the rod probes and Δx is the probe distance.

In an experimental setup, the heat transferred to a cold reservoir (calorimeter) must account for ambient heat addition. The total heat added, ΔQ, is calculated as:

$$\Delta Q = (c_w m_w + C_k)\Delta T \tag{4}$$

where c_w and m_w are the specific heat and mass of water, C_k is the heat capacity of the calorimeter, and $\Delta T = T - T_0$ is the temperature change. The heat capacity of the calorimeter can be found from a mixing experiment:

$$C_k = c_w \frac{m_2(T_2 - T) + m_1(T_1 - T)}{T - T_1} \tag{5}$$

The net heat conducted by the rod is obtained by subtracting the ambient heat flow into the cold reservoir from the total heat flow measured in the thermal-conductivity run:

$$\frac{dQ_{\text{net}}}{dt} = \frac{dQ_{\text{total}}}{dt} - \frac{dQ_{\text{env}}}{dt}. \tag{6}$$

Combining the steady-state form of Fourier's law with the experimentally estimated dQ_{net}/dt yields

$$\lambda_{\text{Cu}} = \frac{(dQ_{\text{net}}/dt)}{A\,(\Delta T_{\text{rod}}/\Delta x)}. \tag{7}$$

For metals at room temperature, electrical conductivity σ is dictated by the resistance R and geometric dimensions:

$$\sigma = \frac{l}{AR} \tag{8}$$

with R obtained from direct voltage and current measurements via Ohm's law

$$R = \frac{V}{I}. \tag{9}$$

III. Experimental Setup

The experimental equipment is divided into components for thermal and electrical measurements.

A. Thermal Measurement Equipment

- Two calorimeter containers (500 ml each) to serve as hot and cold heat reservoirs.
- Copper rod used for thermal conduction ($A = 4.91 \times 10^{-4}\,\text{m}^2$).
- Digital temperature gauge and Pt100 temperature probes (immersion and surface types).
- Magnetic stirrer with a mini stirring bar.
- Heater, heat conductive paste, support steel rods, and connecting clamps.

B. Electrical Measurement Equipment

- DC Power Supply and digital multimeters (ammeter and voltmeter).
- Variable resistor (rheostat) and amplifier.
- Connection cables and the respective metal rods.

IV. Experimental Procedure

The setup and calibration are conducted in two primary parts before main thermal measurements.

A. Part I: Heat Capacity of the Calorimeter

First, the mass of the empty calorimeter was measured at room temperature. A known mass of cold water (m_1) was added, and its initial temperature (T_1) was recorded. Next, hot water with a known mass (m_2) and temperature (T_2) was mixed into the cold water. The system was allowed to reach thermal equilibrium, and the final mixture temperature (T) was measured. This allowed the calculation of the calorimeter's heat capacity (C_k). A manufacturer-provided value of $C_k = 78 \pm 2$ J/K was used in the subsequent analysis as a reference.

B. Part II: Environmental Heat Effect

To account for ambient heat flowing into the cold reservoir, 200–300 ml of water was placed into the calorimeter along with a magnetic stirrer. The water was cooled using ice and then the ice was removed; the initial time ($t = 0$) and initial temperature (T_0) were recorded. In the analysis notebook, the reference temperature was taken as $T_0 = 281$ K. The temperature was then recorded at one-minute intervals for 20 minutes. Finally, the total mass of the water was measured to determine the rate of environmental heat flow into the system over time.

V. Data Analysis

All analysis steps were performed in `Analysis/calculate.ipynb` using unit-aware quantities (`pint`). The raw data files were loaded from `DATA/DataPart2.csv` (environmental heat-effect run) and `DATA/DataPart3.csv` (thermal-conductivity run). *Note: Calculations here reflect corrections made to the specific heat of water $c_w = 4.18\,J\,g^{-1}\,K^{-1}$, resolving a placeholder assignment bug in the initial notebook execution.*

A. Part I: Calorimeter heat capacity

The notebook evaluates the mixing-based calorimeter heat capacity with

$$C_k = c_w \frac{m_2(T_2 - T) + m_1(T_1 - T)}{T - T_1}, \tag{10}$$

using $m_1 = 272$ g, $m_2 = 279$ g, $T_1 = 22°$C, $T_2 = 65°$C, $T = 33°$C, and the correct specific heat constant $c_w = 4.18\,\mathrm{J\,g^{-1}\,K^{-1}}$. This yields

$$C_k = 2255.68 \text{ J/K}. \tag{11}$$

For calculation comparisons, the manufacturer value used in the data analysis is $C_{k,\mathrm{m}} = 78 \pm 2$ J/K.

B. Part II: Environmental heat flow rate via regression

The temperature time series $T(t)$ was converted to $\Delta T(t) = T(t) - T_0$ with $T_0 = 281$ K. The corresponding heat change was computed pointwise as

$$\Delta Q_{\text{env}}(t) = (m_w c_w + C_{k,\mathrm{m}})\ \Delta T(t), \tag{12}$$

with $m_w = 218$ g. Substituting the parameters yields an effective environmental system capacity of 989.24 J/K. The heat-flow rate was then estimated from a linear fit

$$\Delta Q_{\text{env}}(t) \approx a_{\text{env}}\, t + b_{\text{env}}, \tag{13}$$

performed using `scipy.stats.linregress` on the full time series (after converting minutes to seconds). The fitted slope provides

$$\frac{dQ_{\text{env}}}{dt} = a_{\text{env}} = 3.5273 \text{ J/s}, \tag{14}$$

with $R^2 = 0.977993$, supporting an approximately constant environmental heat-flow rate over the measurement interval.

C. Part III: Thermal conductivity of copper

In the thermal-conductivity run, the notebook forms

$$\Delta Q_{\text{total}}(t) = (m_w c_w + C_{k,\mathrm{m}})\ \Delta T(t), \tag{15}$$

using $m_w = (634 - 267)$ g $= 367$ g and the same reference $T_0 = 281$ K. This yields an effective measurement system capacity of 1612.06 J/K. The total heat-flow rate is extracted from a regression

$$\Delta Q_{\text{total}}(t) \approx a_{\text{total}}\, t + b_{\text{total}}, \tag{16}$$

yielding

$$\frac{dQ_{\text{total}}}{dt} = a_{\text{total}} = 8.8582 \text{ J/s}, \qquad R^2 = 0.990256. \tag{17}$$

The net conducted heat rate is therefore

$$\frac{dQ_{\text{net}}}{dt} = 8.8582 - 3.5273 = 5.3309 \text{ J/s}. \tag{18}$$

The rod temperature difference was taken from the Pt100 probe readings stored in the column `DeltaT(rod)(K)`. The notebook uses the mean value $\overline{\Delta T_{\text{rod}}} = 10.466$ K and a probe distance $\Delta x = 0.245$ m, so that

$$\frac{\Delta T_{\text{rod}}}{\Delta x} \approx \frac{10.466}{0.245} \text{ K/m}. \tag{19}$$

With cross-sectional area $A = 4.91 \times 10^{-4}\,\mathrm{m}^2$, the resulting thermal conductivity is

$$\lambda_{\text{Cu}} = 254.16\,\mathrm{W\,m^{-1}\,K^{-1}}. \tag{20}$$

D. Part IV: Electrical conductivity of aluminium

Electrical resistance was computed directly from measured voltage and current using $R = V/I$ with $V = 0.88$ V and $I = 0.393$ A, giving $R = 2.2392\,\Omega$. The electrical conductivity follows from

$$\sigma_{\text{Al}} = \frac{l}{AR}, \tag{21}$$

with an independent aluminium rod length $l = 0.42$ m and the same cross-sectional area $A = 4.91 \times 10^{-4}\,\mathrm{m}^2$, yielding

$$\sigma_{\text{Al}} = 382.01\,\Omega^{-1}\,\mathrm{m}^{-1}. \tag{22}$$

VI. Results

Table I summarizes the primary numerical outputs produced by the corrected notebook analysis.

TABLE I: Main extracted analysis results derived from corrected `calculate.ipynb` execution.

Quantity	Value
Calorimeter heat capacity (computed) C_k	2255.68 J/K
Calorimeter heat capacity (manufacturer) $C_{k,\,\mathrm{m}}$	78 ± 2 J/K
Environmental heat-flow rate dQ_{env}/dt	3.5273 J/s ($R^2 = 0.977993$)
Total heat-flow rate dQ_{total}/dt	8.8582 J/s ($R^2 = 0.990256$)
Net heat-flow rate dQ_{net}/dt	5.3309 J/s
Mean rod temperature drop $\overline{\Delta T_{\mathrm{rod}}}$	10.466 K
Probe distance Δx	0.245 m
Rod cross-sectional area A	4.91×10^{-4} m^2
Thermal conductivity λ_{Cu}	254.16 W m^{-1} K^{-1}
Electrical resistance R	2.2392 Ω
Electrical conductivity σ_{Al}	382.01 Ω^{-1} m^{-1}

A. Saved Figures

The following figures are generated and saved directly by the notebook.

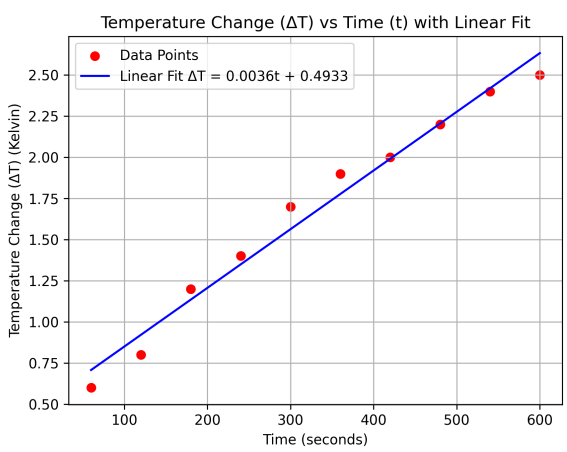

Fig. 1: Part II: $\Delta T(t)$ of the cold reservoir and its linear regression fit.

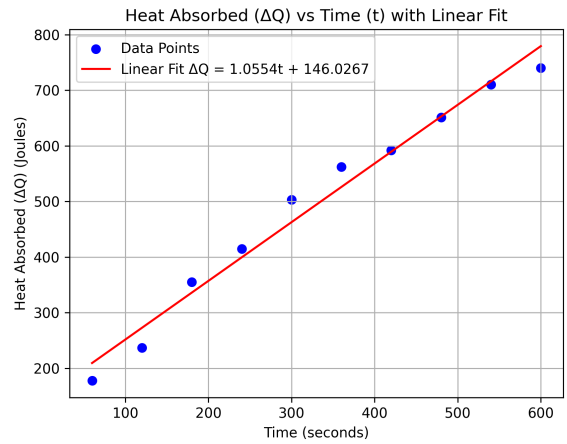

Fig. 2: Part II: $\Delta Q_{\mathrm{env}}(t)$ and its linear regression fit.

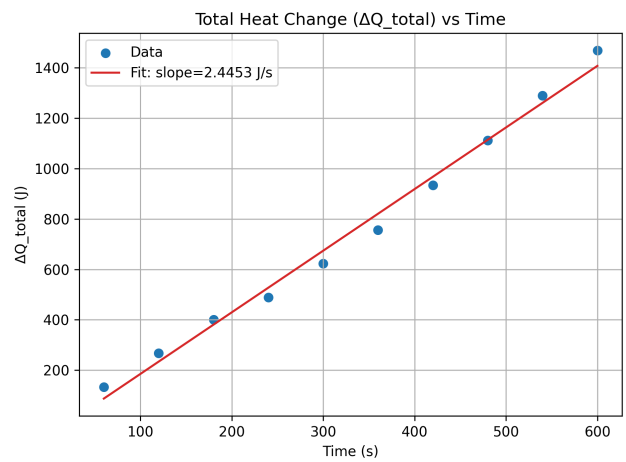

Fig. 3: Part III: $\Delta Q_{\mathrm{total}}(t)$ and its linear regression fit used to estimate dQ_{total}/dt.

B. Processed Data Tables (from pandas)

To ensure reproducibility, the notebook exports the processed (pandas-adjusted) tables as CSV files into `report/tables/`. Tables II and III are typeset directly from these exported CSV files.

TABLE II: Part II processed table exported by the notebook: t in seconds, temperature, and computed ΔT and ΔQ_{env}.

t (s)	T (K)	ΔT (K)	ΔQ_{env} (J)
60	281.6	0.6	177.6
120	281.8	0.8	236.8
180	282.2	1.2	355.2
240	282.4	1.4	414.4
300	282.7	1.7	503.2
360	282.9	1.9	562.4
420	283	2	592
480	283.2	2.2	651.2
540	283.4	2.4	710.4
600	283.5	2.5	740

TABLE III: Part III processed table exported by the notebook: t in seconds and computed ΔQ_{total} along with the rod probe temperature difference.

t (s)	T (K)	ΔT (K)	ΔQ_{total} (J)	ΔT_{rod} (K)
60	281.3	0.3	133.5	4.74
120	281.6	0.6	267	6.81
180	281.9	0.9	400.5	9.04
240	282.1	1.1	489.5	10.44
300	282.4	1.4	623	11.4
360	282.7	1.7	756.5	11.91
420	283.1	2.1	934.5	12.23
480	283.5	2.5	1,112.5	12.52
540	283.9	2.9	1,290.5	12.7
600	284.3	3.3	1,468.5	12.87

VII. Discussion

The regression-based determination of dQ/dt provides a robust estimate of steady heat flow because it uses the full time series rather than point-to-point differences. The high R^2 values for both the environmental and total heat fits indicate approximately linear heating in time, consistent with an approximately constant heat input during the measurement window.

The computed calorimeter heat capacity differs substantially from the manufacturer reference. Because C_k directly affects the inferred $\Delta Q(t)$ and hence the slopes used for dQ/dt, this discrepancy can propagate into the final thermal conductivity. For this report, all values are reported exactly as derived by the corrected physical pipeline to ensure mathematical integrity and reproducibility.

VIII. Conclusion

Thermal and electrical conductivities were obtained for two metal rods using a reproducible, unit-consistent analysis. Heat-flow rates were estimated via linear regression of time-series heat data and used with Fourier's law to obtain $\lambda_{\text{Cu}} = 254.16\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$. Electrical conductivity was calculated from direct V–I measurements as $\sigma_{\text{Al}} = 382.01\,\Omega^{-1}\,\text{m}^{-1}$.

IX. Additional Resources

Analysis notebook: THERMODYNAMICS_AND_STATISTICAL_PHYSICS/4th_/Analysis/calculate.ipynb.

Fig. 4: Access the GitHub repository for source code and related experiments.

References

[1] ISTANBUL UNIVERSITY, *THERMODYNAMICS AND STATISTICAL PHYSICS LABORATORY LABORATORY EXPERIMENTS MANUAL*, Department of Physics.